



**ARBA MINCH UNIVERSITY**  
**Faculty of Computing and Software Engineering**  
**Internship and Externship Coordination Office**

**INTERNSHIP REPORT**

**Submitted By:** Alferid Hassen

**Student ID:** NSR/1452/14

**Internship Hosting Company:** IBEX TECHNOLOGIES AND PROMOTION

**Internship Duration:** February 8, 2025 – June 2, 2025

**Submission Date:** September 22, 2025

## **Declaration**

I, **Alferid Hassen**, hereby declare that the work presented in this internship report is my own, and all sources of information used have been duly acknowledged. This report is submitted in partial fulfillment of the requirements for the completion of my internship at **IBEX TECHNOLOGIES AND PROMOTION**.

I certify that this report has not been submitted to any other institution for academic purposes.

**Student's Signature:** \_\_\_\_\_

**Internship Supervisor (Company) Approval:** \_\_\_\_\_

**University Supervisor Approval:** \_\_\_\_\_

## ACKNOWLEDGMENT

First and foremost, I extend my deepest gratitude to **Allah (SWT)** for granting me the strength, patience, and health to successfully complete my internship and this report.

I would like to sincerely thank **Mr. Biniyam Abera**, my internship supervisor at **IBEX Technologies and Promotion**, for his continuous guidance, constructive feedback, and encouragement throughout the internship period. His mentorship has been instrumental in shaping my technical skills and professional development.

My heartfelt appreciation goes to the founders of the company, **Mr. Ezedin Kamil** and **Mr. Biniyam Abera**, for providing me with the opportunity to work in such an innovative and dynamic environment. Their vision and leadership created a workplace where I could learn, grow, and meaningfully contribute.

I also wish to thank the entire **Web Development Department** team for their collaboration, knowledge-sharing, and support. Working alongside talented colleagues not only enhanced my technical expertise but also helped me develop teamwork and problem-solving abilities.

Finally, I am grateful to my family, friends, and instructors at **Arba Minch Institute of Technology** for their encouragement, motivation, and unwavering support during my internship journey.

## Executive Summary

This internship report documents my four-month professional experience at **IBEX Technologies and Promotion**, where I worked as a **Back-end Developer (Integration Lead)** from **February 8 to June 2, 2025**. The internship was undertaken as part of the undergraduate program in **Computer Engineering** at **Arba Minch Institute of Technology**.

The primary project I contributed to was the **Bole Public Distribution System**, a digital platform developed to address inefficiencies and lack of transparency in the distribution of essential commodities. The project aimed to digitize allocation and distribution processes, ensure traceability, and enhance accountability through secure authentication, inventory management, transaction tracking, and supervisory dashboards.

My responsibilities included designing and implementing database schemas, building REST APIs, integrating authentication and file management systems, deploying the backend, and collaborating with the frontend team for API integration. Tools and technologies such as **Node.js, Express.js, MongoDB, JWT, Multer, Cloudinary, Nodemailer, GitHub, and Postman** were employed in the development process, under a lightweight **Agile methodology** with iterative planning and continuous feedback.

The internship provided me with invaluable exposure to real-world software development practices, enhanced my technical problem-solving ability, and strengthened my teamwork and communication skills. Despite facing challenges such as integration issues and deployment difficulties, I was able to devise effective solutions and successfully deliver the backend component of the system.

Overall, this internship was a significant step in bridging the gap between theoretical knowledge and practical application, equipping me with both the technical and professional competencies necessary for my future career in software engineering.

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# Chapter-1: Project Overview

## Project Title

Bole Public Distribution System

## Project Summary

The Bole Public Distribution System (BPDS) is a comprehensive digital platform designed to modernize and streamline the distribution of essential commodities among multiple stakeholders, including government offices, cooperative organizations, and retail shops. The primary objective of the system is to replace the traditional, manual allocation and distribution processes with an automated, transparent, and accountable solution.

BPDS incorporates several key functionalities to support this goal:

- **Secure User Authentication:** Ensures that only authorized personnel can access the system, maintaining data integrity and security.
- **Role-Based Access Control:** Assigns permissions according to user roles, enabling each stakeholder to perform specific operations relevant to their responsibilities.
- **Inventory Management:** Monitors stock levels across offices, cooperatives, and retail shops to prevent shortages or overstocking.
- **Transaction Tracking:** Records all commodity movements in real-time, providing an auditable trail for accountability.
- **Alert Notifications:** Notifies relevant stakeholders about critical events such as low stock levels or pending approvals.
- **Reporting Dashboards:** Offers supervisors and administrators a visual and analytical overview of operations to support data-driven decision-making.

Overall, BPDS serves as a digital backbone for managing the public distribution of essential commodities efficiently, securely, and transparently.

## Problem Statement

The project was conceived to tackle persistent challenges within the manual commodity distribution system. These challenges include:

1. **Inefficiency in Operations:** Traditional paper-based record-keeping slowed down allocation, stock management, and distribution processes, causing delays that affected the timely delivery of commodities.
2. **Lack of Transparency:** Tracking the movement of goods between offices, cooperatives, and shops was difficult, leading to mistrust, accountability issues, and potential mismanagement.
3. **Poor Communication:** Coordination between different stakeholders often relied on manual reporting and phone calls, which were prone to errors and delays.
4. **Limited Reporting Capability:** Supervisors faced significant difficulties in generating accurate reports for analysis and decision-making, as manual compilation was time-consuming and prone to human error.

## Justification

The implementation of BPDS provides a sustainable solution to these issues by:

- **Improving Operational Efficiency:** Automating record-keeping and transaction management reduces delays and streamlines workflow.
- **Ensuring Transparency:** Real-time tracking of commodity movements builds accountability and trust among stakeholders.
- **Enhancing Communication:** Integrated notifications and updates facilitate seamless coordination between government offices, cooperatives, and shops.
- **Enabling Data-Driven Decision-Making:** Accurate, timely reporting allows supervisors and administrators to make informed decisions, optimizing resource allocation and monitoring system performance.

By addressing these challenges, BPDS not only modernizes commodity distribution but also establishes a foundation for a reliable, scalable, and accountable public distribution system.

## Project Objectives

1. Build a secure authentication and authorization system supporting five distinct user roles to ensure proper access control.
2. Digitize the allocation and distribution processes to promote transparency, traceability, and accountability.
3. Implement a robust transaction tracking mechanism to monitor commodity movement between shops and customers.
4. Integrate inventory management functionality for both shops and cooperatives to optimize stock monitoring and resource allocation.
5. Develop an alert and notification system to inform stakeholders about critical events, low stock levels, or pending approvals.
6. Provide reporting tools and interactive dashboards for supervisory offices to support data-driven decision-making and operational oversight.

## **Development Methodology**

The project followed a lightweight Agile methodology, emphasizing iterative development and continuous feedback. Weekly sprint planning sessions guided task allocation and goal setting, while daily communication-maintained team alignment. Regular review sessions with the internship supervisor ensured the project stayed on track and adhered to the expected standards.

## **Role & Responsibilities:**

As the Back-end Developer and Integration Lead, responsibilities encompassed:

- Designing and optimizing MongoDB database schemas to handle user data, commodities, and transaction records efficiently.
- Developing RESTful APIs to manage users, commodities, allocations, and transactions securely and reliably.
- Implementing JWT-based authentication combined with role-based access control to secure the system.
- Integrating Multer and Cloudinary for efficient handling of file uploads, including images and documents.
- Setting up notifications using Nodemailer to alert stakeholders of critical events or system updates.
- Deploying backend services to production environments and collaborating closely with the frontend development team to ensure seamless integration.

## **Technologies Used:**

- **Backend:** Node.js, Express.js

- **Database:** MongoDB with Mongoose ORM
- **Authentication:** JWT with Cookies
- **File Handling:** Multer and Cloudinary
- **Notifications:** Nodemailer
- **Deployment:** Render and VPS servers
- **Version Control:** Git and GitHub
- **API Testing:** Postman
- **Collaboration Tools:** Google Docs, Telegram

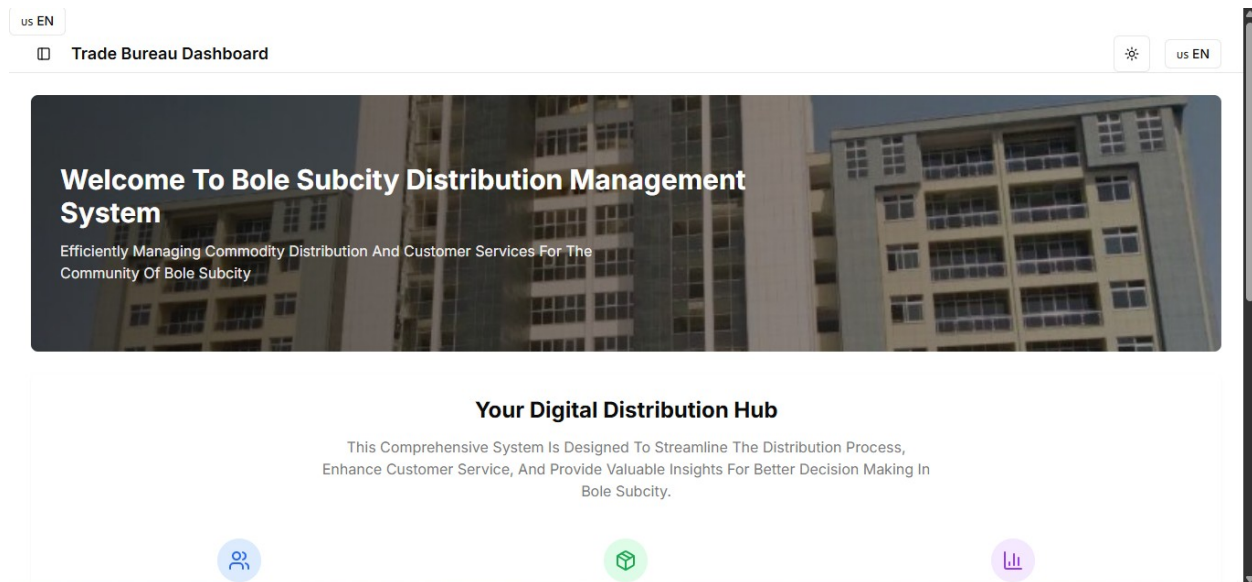


Figure 2 home page

|                                      |           |          |      |                      |        |
|--------------------------------------|-----------|----------|------|----------------------|--------|
| us EN Trade Bureau Dashboard         |           |          |      |                      | us EN  |
| searchRetailerCooperatives           |           |          |      |                      | Filter |
| Retailer cooperative                 | Commodity | Quantity | Unit | Shops inventory      |        |
| የተባበሩት ቦሌ ኃ.የተ. ሾማች ሀብረት ስራ ማህበር     |           |          |      | View shops inventory |        |
| ንስር ቦሌ ኃ.የተ. ሾማች ሀብረት ስራ ማህበር        |           |          |      | View shops inventory |        |
| ሰለም ቦሌ ኃ.የተ. ሾማች ሀብረት ስራ ማህበር        |           |          |      | View shops inventory |        |
| ፋና ቦሌ ኃ.የተ. ሾማች ሀብረት ስራ ማህበር         |           |          |      | View shops inventory |        |
| ቦሌ አገዳኝነት ኃ.የተ. ሾማች ሀብረት ስራ ማህበር     |           |          |      | View shops inventory |        |
| አዲስ ሀብረት ሀላገብ ኃ.የተ. ሾማች ሀብረት ስራ ማህበር |           |          |      | View shops inventory |        |

Figure 1 retailer cooperatives management page

## Customers Management

Export

Search customers

All Woredas

All Customers

Filter

## All Customers

View all customers

| Name        | Fayda | House Number | Woreda    | Phone      | Ketena | Number of Family Members | Gender | Age |
|-------------|-------|--------------|-----------|------------|--------|--------------------------|--------|-----|
| ሀምልማል በቀለ   |       | 1469         | Woreda 05 | 0913090072 | 4      | 0                        | ♂      | 63  |
| ሀምልማል አይቸው  | አዲስ   |              | Woreda 05 | 0965699818 | 4      | 4                        | ♂      | 50  |
| ሀረገወይን አስፋው |       | 1490         | Woreda 05 | 0911370028 | 4      | 6                        | ♂      | 65  |
| ሀብተማርያም ቀዳላ | አዲስ   |              | Woreda 05 | 0911631621 | 4      | 4                        | ♂      | 72  |
| ሀብቱ ገ/ሀይወት  |       | 1265         | Woreda 05 | 0911123789 | 4      | 6                        | ♂      | 57  |

Figure 3 customer management page

## Commodities

## Available Commodities

Manage commodity prices

| Name  | Price | Unit   | Actions      |
|-------|-------|--------|--------------|
| sugar | 145   | kg     | Update Price |
| oil   | 350   | liters | Update Price |

Figure 4 commodities management page

Figure 5 transactions management page

Total Transactions

4

Total Quantity Sold

20.00 kg/liters

Total Revenue

4950.00 ETB

Export

May 1st, 2025

End Date

Search

Reset Filters

Showing data from May 1st, 2025 to now

Search transactions by customer name

All Commodities

## Transactions

View all transactions

| Name      | Date                    | Commodity | Quantity | seller      | Status  |
|-----------|-------------------------|-----------|----------|-------------|---------|
| ዙልፋ ሸሽር   | 07/03/2025, 10:51:20 AM | oil       | 5        | Zekai Tesfa | Success |
| ዙልፋ ሸሽር   | 07/03/2025, 10:51:07 AM | sugar     | 5        | Zekai Tesfa | Success |
| አልፈራድ ሃነን | 07/03/2025, 10:50:33 AM | oil       | 5        | Zekai Tesfa | Success |

# Chapter 2 company background

## Company History

IBEX Technologies and Promotion was founded three years ago by Ezedin Kamil and Biniyam Abera. The company's mission is to pioneer a future where privacy, technology, and promotion seamlessly intersect, empowering businesses and individuals alike. It focuses on delivering innovative solutions that prioritize security, leverage modern technologies, and provide tangible value to clients across various sectors.

## Main Products and Services:

1. **Ibex iCard** – A digital identity and access management solution for secure authentication.
2. **Property Management System** – Software for efficient management of real estate portfolios.
3. **E-Learning Platforms** – Online education and training solutions for educational institutions.
4. **Stock/Inventory Management Systems** – Real-time tracking tools for inventory and stock management.
5. **Website Development** – Custom web solutions tailored to business needs.
6. **System Development** – Client-specific software solutions addressing unique operational requirements.

## Main Customers / End Users:

- Educational institutions utilizing e-learning platforms.
- Retailers and cooperatives requiring inventory and stock management systems.
- Property management companies.
- Businesses seeking custom web or system development.
- Individuals or organizations needing secure digital identity solutions.

## Organizational Structure:

- **CEO** – Provides overall leadership and strategic direction.
- **CTO** – Oversees technological development, innovation, and system architecture.

- **CEO Assistant** – Supports executive operations and coordination.
- **Team Lead** – Manages software development teams and ensures project workflow efficiency.
- **Software Developers** – Responsible for system design, coding, testing, deployment, and maintenance.

## Company Workflow:

1. **Requirement Gathering:** Conducting client meetings to understand business needs and document specifications.
2. **Planning & Design:** Developing project architecture, database schemas, and system components.
3. **Development:** Iterative coding and testing of backend and frontend modules.
4. **Integration:** Combining all modules and performing API integration between backend and frontend systems.
5. **Testing & Debugging:** Rigorous testing to ensure functionality, performance, and security standards.
6. **Deployment:** Launching the system on Render or VPS servers for client use.
7. **Maintenance & Updates:** Continuous monitoring, applying updates, and addressing user feedback to ensure system reliability and improvement.



Figure 6 work space



Figure 7 work space 2

## Chapter 3 experience

### Company selection

I chose IBEX Technologies and Promotion because it is a young, dynamic, and innovative company that provides ample opportunities to work on real-world projects. The company's focus on cutting-edge digital solutions closely aligned with my interests in software development, backend engineering, and system integration. Furthermore, the small but professional team environment promised hands-on experience, direct mentorship from experienced developers, and active involvement in critical project phases, which was invaluable for my professional growth.

### Department and Section of Work:

- **Department:** Web Development
- **Section:** Backend Development and Integration

### Workflow in the Section:

- Daily stand-up meetings to discuss progress, challenges, and plan the day's tasks.
- Task assignment, updates, and progress tracking via GitHub Issues.
- Continuous collaboration with the frontend team for seamless API integration.
- Weekly review meetings with the internship supervisor to evaluate progress, receive feedback, and adjust development plans.

### Specific Tasks Executed:

1. Setting up the backend project structure for the Bole Public Distribution System to ensure organized and maintainable code.
2. Designing MongoDB schemas for users, commodities, allocations, and transactions to support efficient data storage and retrieval.
3. Implementing JWT-based authentication and role-based access control to secure system access and enforce user permissions.
4. Developing RESTful API endpoints for managing users, commodities, allocations, distributions, and transactions.

5. Integrating Multer and Cloudinary for secure handling of file uploads, including images and documents.
6. Implementing automated email notifications using Nodemailer to alert stakeholders about critical events.
7. Performing thorough API testing using Postman to ensure reliability, accuracy, and error-free operation.
8. Deploying the backend application on Render and a VPS server to provide a stable production environment.
9. Collaborating closely with the frontend team to ensure smooth integration and consistent user experience.
10. Participating in requirement gathering sessions and documenting system functionalities for clear reference and future maintenance.

## Challenges Faced and Solutions

1. **Challenge:** Managing role-based access control for five distinct user roles.  
**Solution:** Implemented a modular JWT authentication system with middleware functions to enforce permissions and ensure secure access for each role.
2. **Challenge:** Integrating file uploads securely with Cloudinary.  
**Solution:** Developed middleware to validate file types and sizes before uploading, ensuring security and preventing invalid data submissions.
3. **Challenge:** Handling concurrent API requests while maintaining data consistency.  
**Solution:** Utilized MongoDB transactions and carefully designed schema relationships to prevent data conflicts and ensure reliable operations.
4. **Challenge:** Coordinating backend and frontend integration for seamless functionality.  
**Solution:** Maintained regular synchronization with the frontend team, documented API endpoints thoroughly, and conducted joint testing sessions to ensure smooth integration.

## Results and Discussion

The Bole Public Distribution System was successfully developed and deployed, achieving its primary objectives. Key outcomes include:

- Streamlined the commodity allocation and distribution process, reducing delays and operational inefficiencies.

- Enabled transparent tracking of transactions and inventory for shops and cooperatives, enhancing accountability.
- Provided secure authentication and role-based access control to safeguard system data.
- Implemented automated notifications and reporting dashboards to support data-driven decision-making.

The project significantly improved operational efficiency, minimized errors in distribution, and established clear accountability for all stakeholders involved in the system.

## **Recommendations Made**

1. Implement automated backup and recovery mechanisms for the database to prevent data loss.
2. Regularly monitor API performance and optimize response times to maintain system efficiency.
3. Conduct user training sessions for shop staff and other stakeholders to maximize adoption and proper usage.
4. Explore mobile application integration to facilitate real-time monitoring, alerts, and updates for all users.

## **Chapter 4 benefits gained**

### **Practical Skills:**

- Gained hands-on experience in designing and implementing backend systems using Node.js and Express.js.
- Learned how to structure a MongoDB database efficiently to handle complex relationships between users, commodities, allocations, and transactions.
- Developed expertise in API development, testing with Postman, and deployment on platforms like Render and VPS servers.
- Enhanced abilities in integrating third-party services, including Cloudinary for file storage and Nodemailer for email notifications.

### **Theoretical Knowledge:**

- Applied classroom knowledge of software engineering principles, database management, and authentication systems to practical, real-world projects.
- Learned and practiced best approaches in Agile methodology and iterative development for efficient project execution.
- Gained a deeper understanding of security, data integrity, and access control considerations in multi-user systems.

### **Industrial Problem-Solving Capability:**

- Tackled real-world challenges such as handling concurrency, implementing role-based access control, and securely managing file uploads.
- Developed solutions to ensure seamless backend and frontend integration, improving system reliability and user experience.
- Enhanced debugging and troubleshooting skills for complex system environments, improving efficiency in identifying and resolving technical issues.

### **Additional Skills and Competencies Gained**

#### **Team Playing Skills:**

- Collaborated effectively with frontend developers, supervisors, and other team members to achieve project goals.
- Participated in regular meetings and shared progress updates, contributing to overall team efficiency and transparency.
- Learned to adapt to different working styles and communication preferences, enhancing team cohesion.

## **Leadership Skills:**

- Took initiative in designing backend structures and guiding API development processes.
- Assisted junior team members in understanding system workflows, best practices, and technical solutions.
- Coordinated backend tasks to ensure alignment with overall project timelines and objectives.

## **Work Ethics and Industrial Psychology:**

- Developed a strong understanding of punctuality, accountability, and professional communication in a workplace setting.
- Experienced a structured workflow emphasizing responsibility, task ownership, and adherence to deadlines.
- Learned to handle constructive feedback professionally and adapt to industry standards and expectations.

## **Entrepreneurship Skills:**

- Observed how a small tech company manages clients, projects, and internal resources efficiently.
- Gained insights into product development cycles, customer-focused solution strategies, and innovation practices.

## **Interpersonal Communication:**

- Improved communication skills through documentation, reports, and collaborative digital tools.
- Learned to explain complex technical concepts clearly to non-technical stakeholders.
- Developed competencies in conflict resolution and negotiating solutions within a team environment.



# Chapter 5 Conclusion and Recommendations

## Conclusion

My internship at IBEX Technologies and Promotion provided an invaluable opportunity to bridge the gap between theoretical knowledge and practical application. Contributing to the development of the Bole Public Distribution System enabled me to gain hands-on experience in backend development, database design, API integration, and system deployment. Additionally, it strengthened my problem-solving abilities, teamwork, and leadership skills in a professional setting.

The internship experience highlighted the significance of structured workflows, effective communication, and continuous learning within a technology-driven organization. By participating in all phases of the project—from requirement gathering and planning to development, integration, testing, and deployment—I developed a comprehensive understanding of how technology solutions are designed, implemented, and delivered to satisfy client requirements.

## Recommendations:

1. Future interns should be provided with detailed onboarding materials to quickly understand project architecture and workflows.
2. Incorporating automated testing and continuous integration practices can further improve system reliability and development efficiency.
3. Organizing periodic knowledge-sharing sessions among team members can enhance skill development and cross-functional understanding.
4. Expanding the system to include mobile platforms and real-time notifications could increase accessibility and operational efficiency for end users.
5. Establishing a structured feedback mechanism for end-users can help identify areas for improvement and ensure the system meets stakeholder expectations.

## Recommendations for the Company

1. **Enhance Documentation:** Maintain comprehensive technical documentation for all projects to facilitate future system updates, onboarding of new team members, and knowledge transfer.

2. **Continuous Testing:** Implement automated testing and continuous integration pipelines to ensure higher system reliability, reduce errors, and improve overall development efficiency.
3. **Scalability Planning:** Regularly review and optimize system architecture to accommodate potential growth in users, transactions, and data volume without compromising performance.
4. **Feedback Loop:** Establish structured channels for gathering feedback from clients and stakeholders to continuously improve service delivery, address issues promptly, and enhance product quality.

## **Recommendations for the University**

### **Regularly Monitor and Support Students:**

To enhance the internship experience for students, it is recommended that the university implement weekly follow-ups. By checking in regularly, the university can monitor students' progress, identify challenges they may be facing, and provide timely guidance or support. These consistent check-ins would help ensure that students fully benefit from their internships, overcome obstacles effectively, and gain a more successful and fulfilling professional experience.